

DATA SCIENCE PROJECT REPORT

Google Play Store

Exploratory Data Analysis

App Ratings, Reviews, Categories & Market Insights

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Executive Summary

This report documents a structured Exploratory Data Analysis (EDA) on the Google Play Store dataset, containing 10,841 app records across 13 attributes. The project explores app ratings, review volumes, category distributions, free vs. paid dynamics, and app-level statistics to derive actionable insights about the Android app ecosystem.

The analysis demonstrates hands-on proficiency in data cleaning, type conversion, aggregation, filtering, and communicating numeric findings — core skills for data science, product analytics, and business intelligence roles.

Project Highlights

Metric	Value
Dataset Size	10,841 rows × 13 columns
Total Categories	34 unique app categories
Average App Rating	4.19 out of 5.0
Total Apps with 5-Star Rating	274 apps

Metric	Value
Free Apps	10,039 (92.6%)
Paid Apps	800 (7.4%)
Most Reviewed App	Facebook — 78,158,306 reviews
Avg. Review Count	~444,112 reviews per app
Astrology-themed Apps	3 apps
Tools Used	Python, Pandas, NumPy

1. Introduction

1.1 Project Background

The Google Play Store is the world's largest Android app marketplace, hosting millions of apps across dozens of categories. Understanding what drives ratings, which categories perform best, and how free vs. paid models compare is critical for app developers, marketers, and product managers.

This project performs a comprehensive EDA on a publicly available Kaggle dataset representing a snapshot of the Play Store. It is designed as a portfolio piece showing the ability to handle real-world messy data, answer meaningful business questions, and derive clear insights.

1.2 Objectives

- Understand the structure, shape, and quality of the Play Store dataset
- Compute statistical summaries of app ratings and review counts
- Identify top-rated categories and most-reviewed apps
- Compare free vs. paid app performance on ratings
- Perform data cleaning on mixed-type columns (Reviews field with '3.0M' anomaly)
- Answer 15 structured business questions from the dataset

1.3 The 15 Business Questions Answered

#	Question	Category
1	Display top 5 rows of the dataset	Data Inspection
2	Check last 3 rows of the dataset	Data Inspection
3	Find shape of dataset (rows & columns)	Data Inspection
4	Get info: dtypes, null counts, memory usage	Data Quality
5	Overall statistics (describe)	Statistics
6	How many apps contain 'Astrology' in title?	Search & Filter

#	Question	Category
7	What is the average app rating?	Statistics
8	How many unique categories exist?	Categorization
9	Which category has the highest average rating?	Aggregation
10	How many apps have a 5-star rating?	Filter
11	What is the average number of reviews?	Statistics + Cleaning
12	How many free vs. paid apps are there?	Segmentation
13	Which app has the maximum reviews?	Max/Lookup
14	Show top 5 apps by review count	Ranking
15	Compare average rating: free vs. paid apps	Comparison

2. Dataset Overview

2.1 Data Source

The dataset is the Google Play Store Apps dataset from Kaggle, representing a web-scraped snapshot of app metadata. It is widely used in data science education for practicing EDA, data cleaning, and aggregation with Pandas.

Repository: github.com/princemaheta2627-glitch/google_playstore_eda

2.2 Dataset Structure

The dataset contains 10,841 rows and 13 columns. Not all columns are fully populated — the Rating column has 9,367 non-null values, indicating ~1,474 missing ratings that require awareness during analysis.

Column	Dtype	Non-Null Count	Description
App	object	10,841	App name / title
Category	object	10,841	App category (e.g., GAME, SOCIAL, FAMILY)
Rating	float64	9,367	Average user rating (1.0–5.0 scale)
Reviews	object → float64	10,841	Number of user reviews (required cleaning)
Size	object	10,841	App size (e.g., '19M', 'Varies with device')
Installs	object	10,841	Install count range (e.g., '1,000,000+')
Type	object	10,841	Free or Paid
Price	object	10,841	Price in USD (0 for free apps)

Column	Dtype	Non-Null Count	Description
Content Rating	object	10,841	Audience (Everyone, Teen, Mature 17+)
Genres	object	10,841	More granular genre classification
Last Updated	object	10,841	Date of last update
Current Ver	object	10,840	Current app version
Android Ver	object	10,838	Minimum Android version required

2.3 Data Quality Issues Found

- Missing Ratings: 1,474 apps (13.6%) have no rating value — handled by skipping nulls in mean calculations
- Reviews Column Type: Stored as object (string) instead of numeric — required type conversion
- Anomalous Review Value: Row 10472 had Reviews = '3.0M' (string with unit suffix) — replaced with 3.0 before casting to float
- Corrupted Row: Row 10472 (app: 'Life Made WI-Fi Touchscreen Photo Frame') had Category = '1.9' — an erroneous row from scraping
- Duplicate App Names: 9,660 unique apps out of 10,841 rows — duplicates exist (different versions/entries)

3. Methodology

3.1 Tools & Libraries

Library	Role	Usage
Python 3	Core language	All scripting and logic
Pandas	Data manipulation	read_csv, groupby, value_counts, filtering, type conversion
NumPy	Numeric support	Imported for array support (np)
Kaggle Notebook	Environment	Jupyter-based cloud execution (no local setup required)

3.2 Analytical Approach

- Data Loading — pd.read_csv() from local Kaggle input
- Initial Inspection — head(), tail(), shape, info(), describe()
- Data Cleaning — Identifying and fixing dtype issues (Reviews column, anomalous row)
- Univariate Analysis — Distributions, counts, means of individual columns

- Aggregation — `groupby()` for category-level rating comparisons
- Filtering — Boolean conditions to isolate segments (5-star, free/paid, Astrology)
- Ranking — `sort_values()`, `value_counts()`, `max()` for top-N analysis

3.3 Key Code Patterns Used

Data Inspection

```
- data = pd.read_csv('4-googleplaystore.csv')
- data.shape          # (10841, 13)
- data.info()         # dtypes, null counts, memory
- data.describe()     # numeric stats
- data.describe(include='all') # includes categorical
```

Search & Filter

```
- data[data['App'].str.contains('Astrology', case=False)] # → 3 apps
- data[data['Rating'] == 5.0]                             # → 274 apps
- data['Type'].value_counts()                             # Free: 10039, Paid: 800
```

Data Cleaning (Reviews Column)

```
- data['Reviews'].dtype          # object — needs fix
- data[data['Reviews'] == '3.0M'] # anomalous row found
- data['Reviews'] = data['Reviews'].replace('3.0M', 3.0)
- data['Reviews'] = data['Reviews'].astype('float')
- data['Reviews'].mean()         # → 444,111.93
```

Aggregation & Ranking

```
- data.groupby('Category')['Rating'].mean().sort_values(ascending=False)
- data['Reviews'].sort_values(ascending=False).head()
- data.groupby('Type')['Rating'].mean()
```

Max / Lookup Operations

```
- data[data['Reviews'] == data['Reviews'].max()]['App'] # → Facebook
- index = data['Reviews'].sort_values(ascending=False).head().index
- data.iloc[index]['App'] # Top 5: Facebook, WhatsApp Messenger
```

4. Key Findings & Insights

4.1 Dataset Overview Statistics

Statistic	Value	Notes
Total Apps	10,841	Including duplicates across versions
Unique Apps	9,660	Distinct app names
Total Categories	34	From ART_AND_DESIGN to WEATHER
Rating Range	1.0 – 5.0 (nominal), 19.0 (data error)	Row 10472 has corrupt rating of 19.0
Mean Rating	4.19	Computed over 9,367 valid ratings
Median Rating	4.30	50th percentile
Std Dev Rating	0.54	Low spread — most apps are well-rated

4.2 App Ratings Analysis

The average app rating of 4.19 out of 5.0 is notably high, reflecting a common positive bias in user-generated ratings — users who rate apps tend to rate ones they like. The low standard deviation (0.54) confirms most apps cluster in the 4.0–4.5 range.

- 274 apps (2.5% of the dataset) have a perfect 5.0 rating — typically low-review-count apps
- One corrupted row has a rating of 19.0 — a data quality issue from web scraping (Category = '1.9')
- Events and Education categories show the highest legitimate average ratings

4.3 Category Analysis

The dataset spans 34 unique categories. The category-level groupby analysis reveals meaningful variation in average ratings:

Rank	Category	Avg. Rating	Insight
1	EVENTS	4.44	Event apps are niche and highly rated by engaged users
2	EDUCATION	4.39	Education apps attract motivated, satisfied users
3	ART_AND_DESIGN	4.36	Creative tools consistently well-reviewed
4	BOOKS_AND_REFERENCE	4.35	Reference apps have loyal, satisfied audiences
5	PERSONALIZATION	4.34	Customization apps enjoy strong user satisfaction
~	DATING	3.97	Lowest legitimate category — high churn and expectations

Rank	Category	Avg. Rating	Insight
~	SOCIAL	4.10	Competitive space; Facebook skews review volume

4.4 Free vs. Paid App Comparison

The vast majority of apps on the Play Store are free:

App Type	Count	Avg. Rating	Observation
Free	10,039	4.19	Dominant model; broad audience; ad/IAP-supported
Paid	800	4.27	Slightly higher rating — self-selected, committed users
Other (error row)	1	19.00	Data quality anomaly, should be excluded from analysis

Paid apps score marginally higher on average (+0.08), consistent with academic findings that paying users are more invested and lenient in reviews, while free apps attract a wider range of casual users.

4.5 Reviews Analysis

The Reviews column required data cleaning before analysis — a critical data engineering skill demonstrated in this project:

- One row contained '3.0M' as a string value instead of a numeric — identified and replaced with 3.0 before float conversion
- Average reviews per app: 444,112 — heavily skewed upward by mega-apps
- Facebook holds the highest review count at 78,158,306
- The top 5 most-reviewed apps are Facebook and WhatsApp Messenger (appearing multiple times due to duplicate/version entries)

Rank	App	Reviews (approx.)	Category
1	Facebook	78,158,306	SOCIAL
2	WhatsApp Messenger	~69,000,000+	COMMUNICATION
3	WhatsApp Messenger (alt entry)	~69,000,000+	COMMUNICATION
4	Facebook (alt entry)	~78,000,000+	SOCIAL
5	WhatsApp Messenger (alt entry)	~69,000,000+	COMMUNICATION

4.6 Niche Findings

- Astrology Apps: Only 3 apps in the dataset have 'Astrology' in the title — a very niche vertical with limited competition
- 5-Star Apps: 274 apps hold a perfect rating — these are typically newer or low-volume apps where a small group of enthusiasts rate them highly
- Most Common App by Name: ROBLOX appears as the most duplicated app name in the dataset (different versions/updates)

5. Technical Skills Demonstrated

5.1 Data Loading & Inspection

- `pd.read_csv()` — file ingestion from Kaggle input path
- `data.shape`, `data.info()`, `data.dtypes` — structural exploration
- `data.describe()` with and without `include='all'` — statistical and categorical profiling
- `data.head()` / `data.tail()` — visual sanity checking of data

5.2 Data Cleaning & Type Handling

- Detecting incorrect dtype: Reviews column stored as object instead of float
- Identifying anomalous string value ('3.0M') using boolean filter
- Using `.replace()` to correct the anomalous value before type casting
- Converting column dtype with `.astype('float')` for numeric operations
- Recognizing a corrupt row (Category = '1.9') as a scraping artifact

5.3 Analysis & Aggregation

- `str.contains()` with `case=False` for case-insensitive text search
- `value_counts()` for frequency distribution of categorical columns
- `groupby().mean().sort_values()` for category-level rating aggregation
- Boolean filtering with single and compound conditions
- `.max()` and comparison-based row lookup for maximum detection
- `sort_values().head()` combined with `.iloc[index]` for top-N ranking

5.4 Python Logic

- Chained Pandas operations for concise, readable analysis
- Inline comments explaining logic (e.g., difference between `unique()` and `nunique()`)
- Assigning transformed columns back to DataFrame (`data['Reviews'] = ...`)

6. Conclusion & Future Work

6.1 Summary of Findings

- The Play Store dataset has 10,841 app entries across 34 categories with 13 attributes
- Average app rating is 4.19 — reflecting positive rating bias common on app stores
- Events and Education are the highest-rated categories; Dating scores lowest legitimately
- 92.6% of apps are free; paid apps earn marginally higher ratings (+0.08 on average)
- Facebook dominates review volume with 78M+ reviews; top 5 review leaders are Facebook and WhatsApp
- The Reviews column required cleaning — a '3.0M' string anomaly was detected and fixed
- 274 apps have perfect 5-star ratings; only 3 apps contain 'Astrology' in their name

6.2 Potential Enhancements

Enhancement	Description	Technique
Sentiment Analysis	Analyze user review text for themes	NLP, VADER, BERT
Price vs. Rating Model	Does price predict quality?	Regression Analysis
Install Prediction	Predict installs from rating/reviews	Random Forest, XGBoost
Time Series	Rating trends by Last Updated date	Pandas DatetimeIndex, Plotly
Interactive Dashboard	Visual exploration of all KPIs	Streamlit or Plotly Dash
Clustering	Group apps by behavior profile	K-Means, t-SNE

6.3 Learning Outcomes

This project reinforced several important data science practices: the importance of inspecting dtypes before aggregation, handling mixed-format columns gracefully, and the value of structured question-driven EDA. Working with a real-world scraped dataset with imperfections (missing ratings, string-encoded numbers, corrupt rows) closely mirrors professional analytics work.

7. Project Information

Project Title	Google Play Store Exploratory Data Analysis
Author	Prince Maheta

GitHub Repository	github.com/princemaheta2627-glitch/google_playstore_eda
Notebook File	4-Google-play-store-eda-project-answers.ipynb
Dataset	Google Play Store Apps — Kaggle (10,841 rows, 13 columns)
Platform	Kaggle Notebook (Jupyter / Python 3)
Languages & Tools	Python 3, Pandas, NumPy
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This report was prepared as part of a data science portfolio.

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